

# Introduction to DATASCI 406

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Computational Methods in Statistics and Data Science (DATASCI 406)

## Course Overview

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# Course Goals

At the end of this course you will be able to

- Trade **computation time** for **analytical effort**
- Connect **mathematical theory** to **computational applications**
- Understand and explain **costs, benefits, and limits** of computational techniques
- Implement key **statistical algorithms** in R
- Create your own **statistical models**
- **Analyze data** using computational techniques

# What is Computational Statistics?

**Statistics:** the application of **probability theory** to real **data**.

Goals:

- Specifying **models of data generation**
- Performing **inference** to connect real data to model
  - **Estimate** model parameters
  - **Test** hypotheses
  - **Predict** future observations
- Quantify **uncertainty** in the inference process.
- Understand **operating characteristics** of tools.

**Computational statistics** uses **substantial amounts of computation** to achieve these goals.

## Two fundamental concepts in computational statistics

**Monte Carlo:** replace a quantity you cannot compute with an average over random draws. If we can get a sample  $X_1, \dots, X_n$ , then

$$E(g(X)) = \int g(x)f(x) dx \approx \frac{1}{n} \sum_{i=1}^n g(X_i)$$

**Optimization:** search for the parameter value that makes something best, when you cannot solve for it algebraically.

E.g., for a complicated density  $f$  with parameter  $\theta$ , what value of  $\theta$  would be most consistent with my sample?

## Four Modules

| Module | Name             | Topic                                        |
|--------|------------------|----------------------------------------------|
| 1      | <b>Integrate</b> | Monte Carlo with the distributions you have  |
| 2      | <b>Generate</b>  | Monte Carlo when you need a new distribution |
| 3      | <b>Resample</b>  | Monte Carlo with only your data              |
| 4      | <b>Fit</b>       | Optimization                                 |

Let's see some examples!

# Module 1: Integrate

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## A density

Consider a **random variable**  $X$  with density  $f$ :

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp \left\{ 1.5x - 0.25x^2 - 2.5966 \right\}$$

How would you find  $E(X) = \int_{-\infty}^{\infty} xf(x) dx$ ?

Nothing here looks like an integral you can do by hand.



## Draw, then average

Suppose I hand you `sampleX(n)`, which returns  $n$  draws from  $f$ .

```
x <- sampleX(100000)
mean(x)
```

```
## [1] 2.999
```

## More complicated integrals

$$E(|X|) = \int_{-\infty}^{\infty} |x| f(x) dx$$

$$\Pr(X > 5) = \int_5^{\infty} f(x) dx$$

$$E(\log(1 + e^X)) = \int_{-\infty}^{\infty} \log(1 + e^x) f(x) dx$$

$$E((1 + X^2)^{-1}) = \int_{-\infty}^{\infty} \frac{f(x)}{1 + x^2} dx$$

## Same code, different $g$

Each is  $E(g(X)) = \int g(x)f(x) dx$  for a different  $g$ .

```
c(EabsX = mean(abs(x)),  
  P_gt_5 = mean(x > 5),  
  Esoft = mean(log1p(exp(x))),  
  Einv = mean(1 / (1 + x^2)))
```

```
##   EabsX  P_gt_5   Esoft   Einv  
## 3.01620 0.07904 3.10634 0.17436
```

Change  $g$ , keep everything else.

## Sampling in R

Every common distribution has an `r` function:

|                    |               |                     |          |
|--------------------|---------------|---------------------|----------|
| <code>rnorm</code> | Normal        | <code>rbinom</code> | Binomial |
| <code>runif</code> | Uniform       | <code>rpois</code>  | Poisson  |
| <code>rexp</code>  | Exponential   | <code>rgamma</code> | Gamma    |
| <code>rt</code>    | Student's $t$ | <code>rbeta</code>  | Beta     |

All take the number of draws first: `rnorm(n, mean, sd)`.

```
sampleX
```

```
## function (n)
```

```
## rnorm(n, mean = 3, sd = sqrt(2))
```

## Functions of samples

A **statistic** is a function of a sample:

$$T = T(X_1, X_2, \dots, X_n)$$

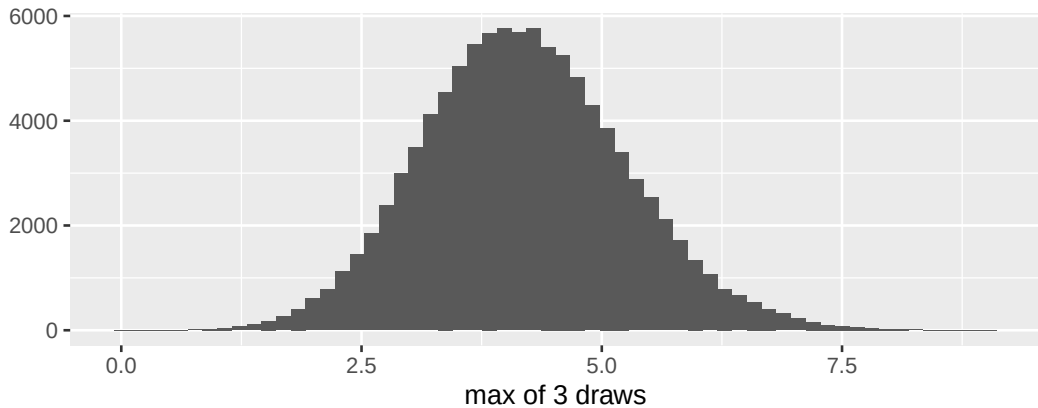
We use statistics to

- Estimate population parameters
- Test hypotheses about populations
- Perform prediction for new observations

A statistic is a random variable, so it has a distribution.

$$T = \max_{i=1,2,3} X_i$$

```
sample_max <- function(n) max(sampleX(n))  
empirical_distrib <- replicate(100000, sample_max(3))
```



## Some Properties

```
mean(empirical_distrib); var(empirical_distrib)
```

```
## [1] 4.195
```

```
## [1] 1.124
```

```
quantile(empirical_distrib, c(0.25, 0.5, 0.75))
```

```
##    25%    50%    75%
```

```
## 3.465 4.157 4.882
```

These are **estimates**. How far off? That is module 1.



## Module 2: Generate

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## A distribution R does not have

The **Pareto** distribution: incomes, city sizes, file sizes. A heavy right tail, so extreme values are far more common than under a Normal.

For shape  $\alpha > 0$  and  $x \geq 1$ :

$$\Pr(X > x) = x^{-\alpha}$$

Therefore it has a **cumulative distributiun function** (CDF)

$$F(x) = 1 - x^{-\alpha}, \quad x \geq 1$$

## The quantile function

Given a probability  $u$ , which  $x$  has  $F(x) = u$ ?

$$u = 1 - x^{-\alpha} \implies x^{-\alpha} = 1 - u \implies F^{-1}(u) = (1 - u)^{-1/\alpha}$$

$F^{-1}(0.5)$  is the median,  $F^{-1}(0.9)$  the 90th percentile.

```
qpareto <- function(u, alpha = 3) (1 - u)^(-1/alpha)
qpareto(c(0.5, 0.9, 0.99))
```

```
## [1] 1.260 2.154 4.642
```

## How to sample? Inversion!

Later we will prove: If  $U \sim \text{Unif}(0, 1)$ , then  $F^{-1}(U)$  has CDF  $F$ .

Algorithm in R:

- Sample  $U$  using `runif`
- Create  $X$  using  $F^{-1}(U)$

## A Pareto sampler in three lines

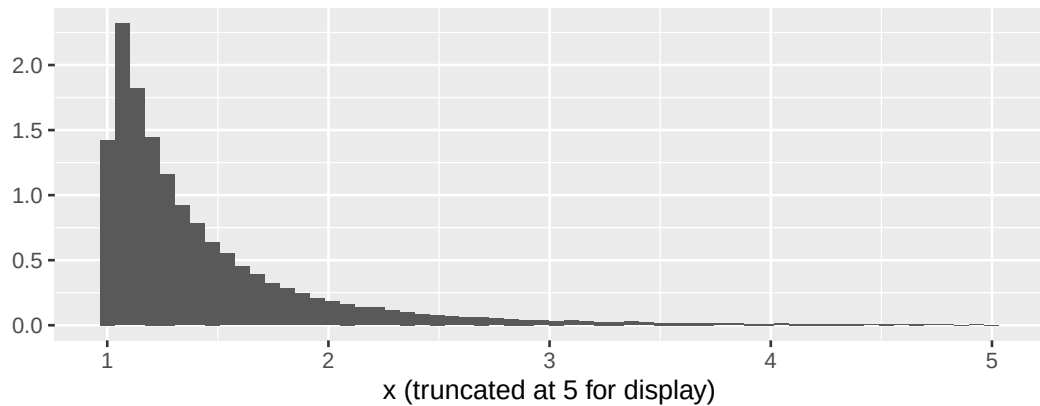
```
rpareto <- function(n, alpha = 3) qpareto(runif(n), alpha)
```

```
x <- rpareto(100000)
```

```
c(mean = mean(x), exact = 3 / 2, min = min(x))
```

```
## mean exact min
```

```
## 1.502 1.500 1.000
```



## Module 3: Resample

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## Unknown distributions

In the previous examples, we had **known distributions** (later we will see these in hypothesis tests and estimating parameters of assumed population distributions).

What about situations where we **don't want to assume a population distribution?**

Idea: Instead of writing down a CDF  $F$ , what if we estimated it?

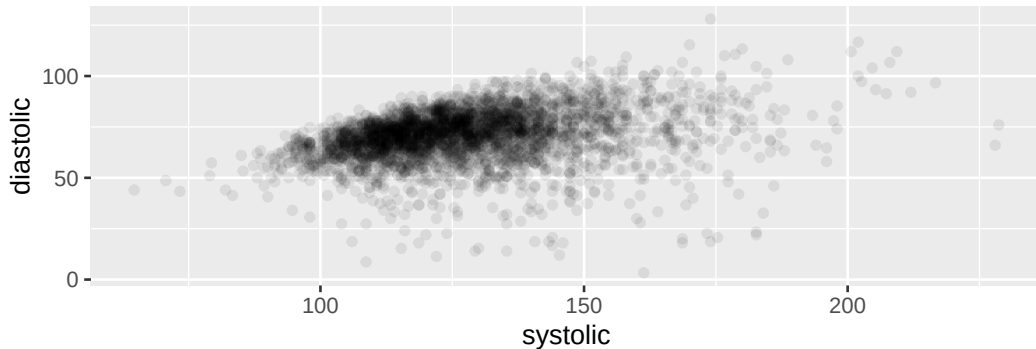


## What distribution is this?

Systolic and diastolic blood pressure, NHANES 2013–14.

```
load("../..data/aspirin.rda")  
dim(combined)
```

```
## [1] 3554    5
```



## No named family fits

```
## systolic diastolic  
##    0.8871    -0.6222
```

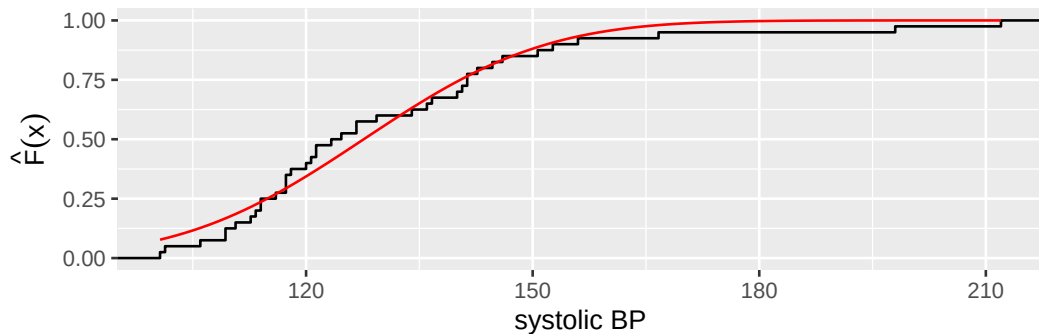
The two margins skew in **opposite directions**, so this is not bivariate Normal. There is also a clump of implausibly low diastolic readings.

We could go hunting for a family that fits. Or we could stop assuming one.

## Estimating the CDF

The **empirical CDF** puts mass  $1/n$  on each observed value:

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\}$$



## Inversion, one module later: The Bootstrap

Module 2: to sample from  $F$ , apply  $F^{-1}$  to a uniform draw.

Now do exactly that to  $\hat{F}$ . Since  $\hat{F}$  puts mass  $1/n$  on each observed point,  $\hat{F}^{-1}(U)$  returns **one of your data points, chosen uniformly at random**.

sampling from  $\hat{F} = \text{sample}(x, \text{replace} = \text{TRUE})$

We call this “bootstrapping.” It is inversion applied to the CDF you estimated.

## Correlation of systolic and diastolic BP

Blood pressure, NHANES 2013–14:

```
theta <- cor(combined$sys_mean, combined$dia_mean)
c(n = nrow(combined), r = theta)
```

```
##           n           r
## 3554.000    0.293
```

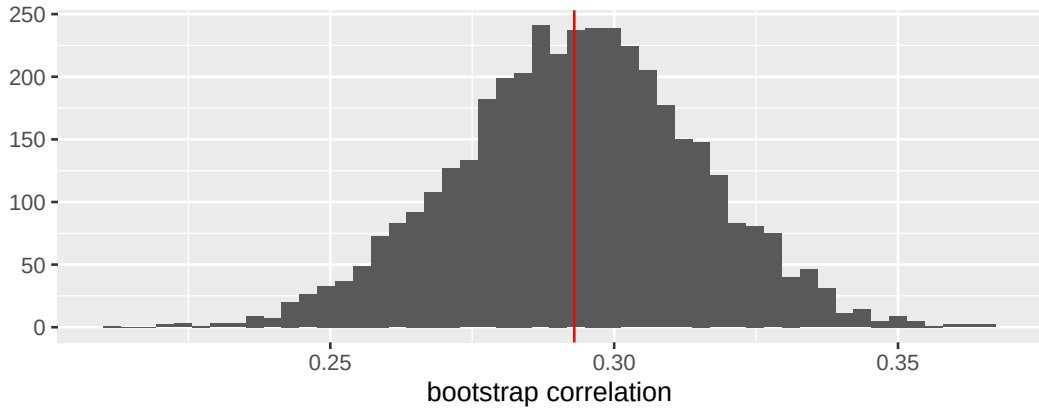
What is the **standard error of that correlation**? What would be a good **95% confidence interval**?

## Bootstrapping for SE, quantiles

Treat the sample as the population and draw from it.

```
set.seed(406)
boot_r <- replicate(4000, {
  i <- sample(nrow(combined), replace = TRUE)  # rows, not values
  cor(combined$sys_mean[i], combined$dia_mean[i])
})
c(se = sd(boot_r), quantile(boot_r, c(0.025, 0.975)))
```

```
##      se    2.5%   97.5%
## 0.02137 0.25020 0.33478
```



## Module 4: Fit

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## Only the maximum survives

Same  $X \sim f$  as before,  $\sigma^2 = 2$  known. We draw  $m = 200$  groups of three, but the record keeps only the largest value in each group:

$$M_j = \max(X_{j1}, X_{j2}, X_{j3}), \quad j = 1, \dots, 200$$

```
set.seed(406)
m <- 200
groups <- matrix(sampleX(m * 3), nrow = m)
M <- apply(groups, 1, max)    # all we get to see
```

Estimate  $\mu$ .

## How to estimate?

Note that  $\mu$  is not the same as  $E(\max X)$ :

```
mean(M)
```

```
## [1] 4.299
```

1. Maximum likelihood estimation
2. Algorithm that converges to the MLE

## Maximum likelihood, briefly

The **likelihood** is the probability of the data we saw, read as a function of the unknown parameter:

$$L(\mu) = \prod_{j=1}^m f_M(M_j; \mu) \qquad \hat{\mu} = \arg \max_{\mu} L(\mu)$$

Pick the  $\mu$  that makes what we observed as likely as possible. More on this in the statistical review.

## Why not just maximize the likelihood?

The max of three draws has density  $3f(x)F(x)^2$ , so

$$\ell(\mu) = \sum_{j=1}^m \left[ \log \phi\left(\frac{M_j - \mu}{\sigma}\right) + 2 \log \Phi\left(\frac{M_j - \mu}{\sigma}\right) \right]$$

Setting  $\ell'(\mu) = 0$  gives

$$\sum_{j=1}^m \frac{M_j - \mu}{\sigma^2} = \frac{2}{\sigma} \sum_{j=1}^m \frac{\phi\left(\frac{M_j - \mu}{\sigma}\right)}{\Phi\left(\frac{M_j - \mu}{\sigma}\right)}$$

$\mu$  is stuck inside  $\Phi$ , which is itself an integral with no closed form. **You cannot solve this for  $\mu$ .**

## What we would do with the full data

With all  $3m$  values in hand, estimating  $\mu$  is a sample mean:

$$\hat{\mu} = \frac{1}{3m} \sum_{j=1}^m \underbrace{\sum_{i=1}^3 X_{ji}}_{S_j}$$

We only need the **group totals**  $S_j$ , not the individual values.

We cannot compute  $S_j$ . But we know one of its three terms and have a bound on the other two.

## Making a reasonable guess

Replace the total we cannot compute with **its expectation given what we saw**.

Given  $M_j$ , one value equals  $M_j$  and the other two are draws from  $f$  conditioned to lie below  $M_j$ :

$$E(S_j | M_j) = M_j + 2E(X | X \leq M_j)$$

For a Normal the conditional mean is closed form:

$$E(X | X \leq a) = \mu - \sigma \frac{\phi\left(\frac{a-\mu}{\sigma}\right)}{\Phi\left(\frac{a-\mu}{\sigma}\right)} < \mu$$

It depends on  $\mu$ , which is what we are trying to find. So guess, then iterate.

## Example

Start from the observed average maximum,  $\mu^{(0)} = 4.30$ , and take the first group, which came back with  $M_j = 5.77$ :

$$E(X \mid X \leq 5.77) = 3.91$$

$$E(S_j \mid M_j) = 5.77 + 2(3.91) = 13.60$$

The true total was  $1.16 + 2.57 + 5.77 = 9.50$ .

One group is badly wrong. The M step averages 200 of them.

## EM: Expectation-Maximization algorithm

Start with a guess  $\mu^{(0)}$ . For  $t = 1, 2, \dots$ :

1. **E**: compute  $E(S_j \mid M_j)$  for every group, using  $\mu^{(t-1)}$ .
2. **M**: set  $\mu^{(t)} = \frac{1}{3m} \sum_j E(S_j \mid M_j)$ .
3. Repeat until it **convergence** (estimate stops moving).

Starting from  $\mu^{(0)} = 4.30$ , one step gives 3.49.



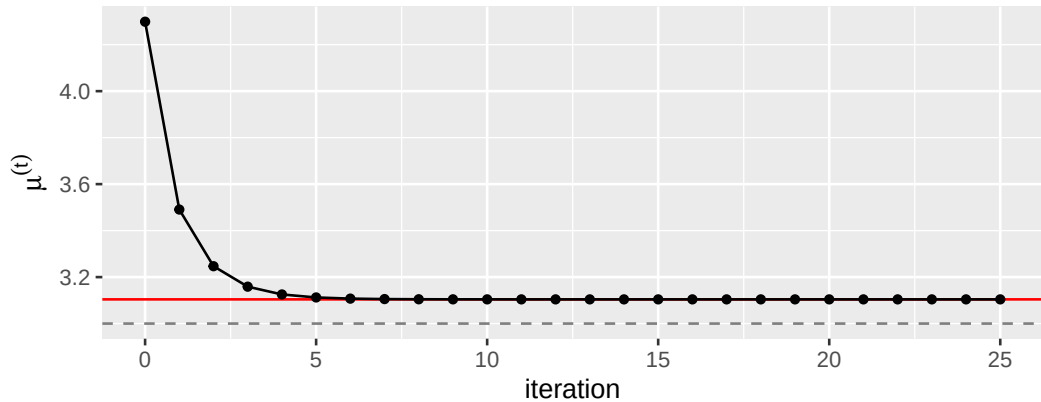
## EM in R

```
sigma <- sqrt(2)
Etrunc <- function(a, mu, s) mu - s * dnorm((a-mu)/s) / pnorm((a-mu)/s)

guess <- mean(M)
for (i in 1:100) {
  e_step <- (M + 2 * Etrunc(M, guess, sigma)) / 3
  m_step <- mean(e_step)
  guess <- m_step
}
c(EM = m_step, naive = mean(M), mu = 3)
```

```
##      EM naive      mu
## 3.104 4.299 3.000
```

# Convergence



Red is the **MLE from the data we actually have**; dashed grey is  $\mu = 3$ .

## What is EM converging to?

Not  $\mu$ , and not the average of the values we lost. EM maximizes the likelihood of the data **we observed**:

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|                                |              |
|--------------------------------|--------------|
| population $\mu$               | 3.000        |
| mean of the 600 hidden values  | 3.063        |
| <b>MLE from the 200 maxima</b> | <b>3.104</b> |
| EM after 25 steps              | 3.104        |

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EM lands on the MLE, exactly. The gap from 3 is **sampling error in 200 maxima**, not a flaw in the algorithm. With more groups the MLE moves toward 3.

## Course Logistics

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**Professor** Mark Fredrickson (<mfredric@umich.edu>)

**GSIs** Ishan Paul (<ishanpl@umich.edu>), Qingyuan Yu  
(<qyyu@umich.edu>)

**Student Hours** TBA, see Canvas

## Prerequisites

- Some programming experience (R, Python, C/C++, Java)
- An understanding of the following terms: Distributions and random variables, random sampling and sampling distributions, population parameters and estimation, hypothesis tests, basic calculus (integrals, derivatives).

No required text, but several **recommended** (particularly first 2). Links to electronic copies available through the library on Canvas.

- Rizzo, Maria L. *Statistical Computing with R*, 2ed
- Wickham, H. & Grolemund, G. *R for Data Science*
- Irizarry, R. A. *Introduction to Data Science*
- Agresti, A. *Foundations of Linear and Generalized Linear Models*
- Gentle, J. E. *Computational Statistics*
- *Handbook of Computational Statistics*. H"rdle, Gentle, and Mori, eds.

All slides and videos will be posted to **Canvas** and **OneNote**.

## Approach

The class is composed of **four modules** and uses a **flipped classroom** approach.

Each module consists of short **check-in quizzes**, a **project**, and an **exam**. There is no separate final exam.

Materials are presented in **video lectures**. Class time is for short **check-in quizzes**, working on examples and the project, and exams.



# Quizzes

- 4–6 multiple choice plus one short answer, every session
- Covers the material you watched **before** class
- **Lowest per module dropped. No makeups.**
- I will reach out to students with testing accommodations.

# Projects

Each module includes a project. The project gives you an opportunity to:

- Implement key statistical algorithms
- Investigate the properties of those algorithms
- Apply these techniques to real-world data

These projects become your notes for your exams. **You will be asked questions on the exams about your project.**

Each module concludes with an exam. Approximate coverage:

- 50%: General conceptual questions, similar to the quizzes.
- 50%: Questions about the results of your own project.

You may bring any **paper** notes you wish. No electronic devices. Held during class periods, with the exception of the Module 4 exam, which uses the scheduled final exam time.

Students with accommodations will be contacted with alternate exam location information.

Each module counts for 25% of your grade. Within each module:

- Exam: 20% of overall grade
- Quizzes: 5% of overall grade, one drop per module

***Permitted and expected*** on projects and all out-of-class work.

***Not permitted*** on quizzes or exams.

Remember: you will be asked about project details. Be prepared to explain coding details, even if you used AI to assist.

## Other Policies

- Course Recordings
- Disability Statement
- Mental Health and Well-Being
- Sexual Misconduct Policy



## Next Session

- No quiz
- Video: statistical review
- Class period: Introduction to R and RMarkdown
- All students: install R and RStudio on laptops, directions on Canvas